METRON

VOLUME IX · ACADEMIC YEAR 2025-2026

A chronicle of the year's accomplishments, initiatives, and triumphs across the EIE family.

STUDENT PLACEMENT OF THE YEAR

Akash Pradeep Kumar

S7 · EIE

Selected for a 6-month paid research internship at NPOL, DRDO Ministry of Defence, Kochi.

NPOL · DRDO

ACADEMIC PATHBREAKER

Fida Fathima E N

S1 · EIE

The first student in FISAT history to clear the Challenge Course in the very first semester.

FIRST-EVER · CHALLENGE COURSE

From the *Editor's* Desk.

This volume is not merely a record; it is a portrait of who we are becoming.

I am delighted to present **METRON IX**, the ninth annual newsletter of the Department of Electronics and Instrumentation Engineering at FISAT. This volume maps the significant developments, achievements, and structural progress attained throughout the 2025-2026 academic year.

EIE has achieved stellar milestones, securing top-tier national awards in mechatronics, winning agricultural hackathons, and earning highly prestigious paid research placements within India's premier defence and aerospace establishments. These accomplishments stand as a direct reflection of the dedication shared by our student body and faculty members.

EDITOR & PUBLICATION TEAM

Mohamed Afnnan S2
A B Navanika S4
Vydhehy T M S6
Annmary Benny S8

FACULTY ADVISOR

Dr. Anish Mathew K

RECOGNITION

Google Student Ambassador 26

Mohamed Afnnan Veettilayil · Chief Editor, METRON IX

ABOUT THE DEPARTMENT

The Department of Electronics and Instrumentation Engineering (EIE) at FISAT delivers comprehensive, multi-disciplinary engineering education and practical design training across electronics, automation, control systems, and smart sensor nodes.

Vision

To emerge as a world-class department in electronics and instrumentation engineering by imparting excellence-focused, interdisciplinary education that molds socially committed and globally competent professionals capable of meeting the evolving demands of industry and research.

Mission

- To provide quality technical education in electronics and instrumentation engineering that enhances employability and meets global industry standards.
- To encourage students to pursue higher education and research, and to become innovative, value-driven professionals with strong ethical foundations.
- To offer state-of-the-art facilities and real-world exposure that prepares students to take on complex engineering challenges.
- To empower students to contribute to technological and societal development, fostering leadership and a sense of social responsibility.

PEOs of the Program

The students of Electronics & Instrumentation Engineering of FISAT will

PEO1 • Design, develop, analyse, and solve engineering problems in core as well as interdisciplinary fields, with social awareness, environmental responsibility, and a commitment to global standards.

PEO2 • Adapt to modern technological trends and evolving industry demands through lifelong learning, research, and innovation, with an orientation towards entrepreneurship and employability.

PEO3 • Practice professional ethics, demonstrate leadership, teamwork, and effective communication skills, and emerge as socially responsible, globally competent professionals with high professional competence.

Program Specific Outcomes

PSO1: Apply the concepts of Electronics and Instrumentation Engineering to design and develop components, systems, and solutions for applications including and not limited to electronics, control systems, process and industrial instrumentation, signal processing, and interdisciplinary areas of engineering, with due consideration for innovation and societal needs.

PSO2: Select, adapt, and apply state-of-the-art hardware and software tools to analyze and solve complex problems in Electrical, Electronics, and Instrumentation Engineering, ensuring solutions that meet industry standards and global practices

Voices from the *top.*

METRON 9 continues to serve as a magnificent mirror documenting EIE's outstanding progress. I congratulate the staff and students on their accomplishments, and compliment the editorial team for crafting this premium newsletter.

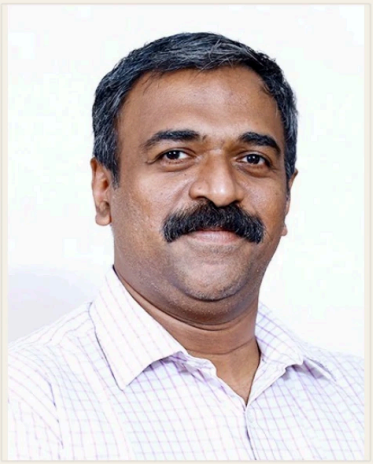
Mr. Shimith P R
CHAIRMAN · FISAT

At FISAT, we view every day and action as a critical learning opportunity. The EIE department has consistently brought laurels to the institution, proving its engineering caliber in global challenges. I wish them the very best.

Dr. Jacob Thomas V
PRINCIPAL · FISAT

This academic cycle has been a landmark phase in EIE's journey toward becoming a Centre of Excellence. From securing joint patents to placing students in top-tier ISRO and DRDO hubs, the dedication of the EIE family is inspiring. Let's keep this momentum going.

Dr. Anil Jhony
HEAD OF DEPT. · EIE



Dr. Anil Johny, Head of the Department of Electronics & Instrumentation Engineering, FISAT.

A YEAR IN BRIEF

From an unprecedented finalist berth at the **International Rover Challenge 2026** to research placements at **NPOL · DRDO** and **VSSC · ISRO**, EIE 202526 has been defined by precision, ambition, and engineering imagination.

Where curiosity *met craft.*

17 SEP 2025 — TECH FEST

AUREATRON 2025

IDEA Club's flagship technical fest brought together Tracker Bot, Guess Your Goal, InnoMark and Soul Mosaic.

HOSTED BY · IDEA Club · EIE Student Association

09 SEP 2025 — WORKSHOP

FPGA Prototyping with Verilog

An advanced technical workshop mapping FPGA prototyping and Verilog logic models, validating digital designs directly on hardware boards.

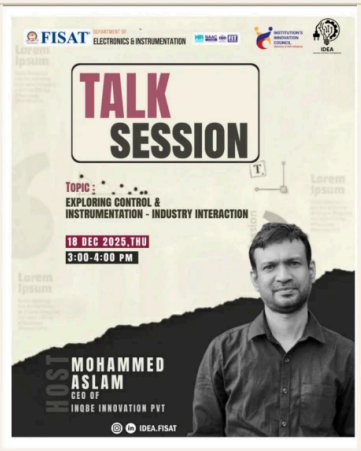
LED BY · Dr. Krishnakumar S & Ms. Jilu George · ECE Dept

23 AUG 2025 — FACULTY DEV.

Outcome-Based Education FDP

A focused programme on *OBE with emphasis on Graduate Attributes & GAPC v4.0*, organised with IQAC and attended by the EIE faculty body.

LED BY · Dr. Baskar S · Dean R&D, Thiagarajar CE



18 DEC 2025 — INDUSTRY VOICE

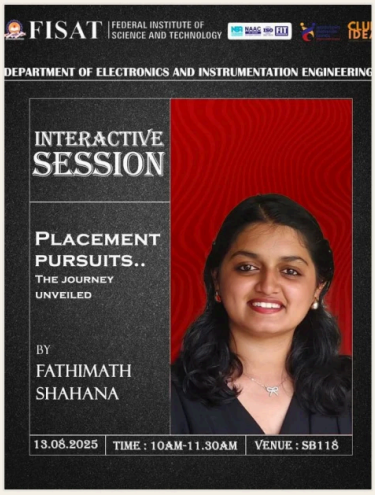
Real-World Control & Instrumentation

A talk session on control and instrumentation practice, connecting classroom theory with industry interaction.

LED BY · Mr. Mohammed Aslam · CEO, INQBE Innovation Pvt Ltd

Bridging campus and *industry.*

MOU SIGNED · 05 NOV 2025



Alumnus-led resume and placement workshop with Reliance Jamnagar Refinery, 13 August 2025.

A renewed pact with NEST Cyber Campus.

The EIE Department renewed its Memorandum of Understanding with NEST Cyber Campus, deepening the existing collaboration around knowledge transfer, student training, and joint technical projects with a sharpened focus on high-power industrial electronics setups.

Nikitha Binu, Partiv A and Annmary Benny at the ISA India Automation Games 2025, MPSTME, Mumbai.

SEVEN FURTHER ENGAGEMENTS ACROSS THE YEAR

13 AUG 2025
Reliance Jamnagar
Alumnus-led mock placement & resume workshop for S7.

19-20 SEP 2025
ISA IAG 2025 · Mumbai
Nikitha Binu, Partiv A & Annmary Benny · ISA IAG, MPSTME.

18 OCT 2025
Oil & Gas Webinar
S7 EIE briefed on flow-control and oil & gas careers.

28 NOV 2025
IIC Regional Meet
Ms. Raji P at the 3rd IIC Regional Meet, Rajagiri.

29 NOV 2025
PALS EDC 360°
Dr. Abi P Mathew & Mr. Tom Anto at the conclave.

12 DEC 2025
Explore 2025 · Naipunnya
Dr. Abi P Mathew as chief guest, Naipunnya School.

18 DEC 2025
Coffee w/ an Edupreneur
Ms. Raji P · Bharata Mata College · IEDC.

INTERNATIONAL ROVER CHALLENGE 2026
MIT Udupi · Karnataka · Grand Finalists

05 SPECIAL FEATURE

Engineering for *another planet.*

38	01	FINALS
STUDENTS	MENTOR	GRAND STAGE
S4 & S6 EIE mentees	Dr. Sreevidya P	28 Jan 02 Feb 2026

A powerhouse team of 38 Semester 6 and Semester 4 EIE students, under the mentorship of **Dr. Sreevidya P**, was shortlisted for the grand finals of the prestigious **International Rover Challenge (IRC-26)** at Manipal Institute of Technology, Udupi. The EIE rover codenamed **DAWN** was custom-engineered to simulate actual planetary surface terrain operations, carrying an advanced **Bio-Detection Module** as its primary system challenge to perform direct organic-trace scans on soil analogs. The project stands as the pinnacle EIE hardware engineering feat of the year, combining robotics, telemetry, and smart sensor instrumentation.

The thirty-eight who *built DAWN.*

A continuous wall of contributors every name a vector of effort behind the rover that earned EIE its place on the international finals stage.

S4 SQUAD · THIRTEEN MENTEES

A P Nimil Nazeer · Nivin Prakash · Darwin T Jose · A B Navanika · Farzeen Fathima N · Ann Maria Benny · Vinay Krishna · Salvion Thomas · Parthip Aji · Shinto Pompson · Karthik P K · Ajwad Ibn Noushad · Bhoumick Narayanan.

S6 SQUAD · TWENTY-FIVE MENTEES

Captain Hazim Karat · Anirudh M · Govind Gopan · Vydhehy T M · Dulfukkar Ali P B · Fahad Yazin · Karthik M D · Emil K Manoj · Devika Rajeev · S. S. Sushobh · Niranjan J Nair · Nibha Prasad · Anjaly Krishna · Saniya Thankachan · Gopika M · Aysha Nourene · Parvathy Shibu · Heera Henry · Fathima Nishida · Aaron Varghese Jacob · Leo Denny · Anerudh M K · Anurag E · Adithyan Ajith · Hareesh S Nair.

THE ENGINEERING BRIEF

Custom-built planetary terrain operations rover with a **Bio-Detection Module** capable of performing organic-trace scans on soil analogs combining robotics, telemetry, and smart sensor instrumentation in a single integrated platform.

THE VENUE & WINDOW

Manipal Institute of Technology, Udupi, Karnataka the grand finals stage of the International Rover Challenge hosted between **28 January and 02 February 2026**. FISAT EIE: among the national shortlisted finalists.

When engineering meets imagination, even another planet begins to feel like home.

IRC-26 CLOSING NOTE

The sophomore *surge.*

Across competitions, hackathons, line-follower arenas, and athletics fields, Semester 4 EIE walked away with a year that doubled as a thesis on consistency.

AgriHack 2025 · hosted by Kerala Agricultural University, Mannuthy.

1ST PLACE · AGRIHACK

A clean sweep at *Mannuthy.*

Salvion Thomas, **Nivin Prakash** and **Farzeen Fathima N** topped the field at the **AgriHack 2025** hosted by Kerala Agricultural University, outpacing competition with an agri-tech solution that fused sensor design with field practicality.

- **1st Place** · Metroticketing · ISA Manipal · Salvion Thomas, Agnes Joela
- **2nd Place** · PLC Programming · ISA Manipal · Nivin Prakash
- **2nd Place** · Idea Pitching · ISA Manipal · Salvion Thomas, Shinto Pompson
- **2nd Place** · CAD Design · ISA Manipal · Karthik P K

ROBOTICS · IDEA PITCHING · SPORT · ACROSS THE YEAR

PATHX · 1ST

Line-Follower at Viswajyothi

Ibrahim K, Amir Shihab, Amaan Shakeer, Izza Muhammed, Aneena Manohar V, Nourin Shanavas.

MBITS · 1ST

Line-Follower at MBITS

Hrithika S, **Mohammed Ajmal**, **Anirudh P** · first place at Mar Baselios IT & Science.

NAVODAYAM · 2ND

Idea Pitching · Chinmaya VV

Adithyan K and **Manal Nivil Ali** secured second place at Chinmaya Vishwavidyapeeth.

ECHO FISAT · 2ND

ISRO Idea Pitching

Kumari Aarthi & **Salvion Thomas** took second place in the ISRO-themed ECHO pitch competition.

The cross-cohort *honor roll.*

Sports achievers across the cohort.

- **Alna V George** · Athletics 4×400 relay · 3rd, KTU athletics meet.
- **Deen Dayal** · Tug of War · 1st, KTU Interzone.
- **A B Navanika** · Tug of War · 2nd, KTU Interzone.
- **Ameen Ahammed** · Football · 1st D Zone; 3rd in KTU Interzone.

Alna V George · 4x400 Relay

Deen Dayal, A B Navanika and Nivedya P S · Tug of War.

Alna V George · Interzone Football.

Ameen Ahammed · D Zone Football.

CREATIVE RECOGNITION

Fashion design wins.

- **Anjima K Baiju** and **Sara Sunny** secured first position in Fashion and Design competitions at IIM Kozhikode, Chinmaya Vishwavidyapeeth, and Mar Athanasius College of Engineering.

Anjima K Baiju and Sara Sunny · Fashion and Design first prize.

Semester six, *in full motion.*

Project expos, national exhibitions, hackathons and trophy-bearing campaigns: a steady cadence of Semester 6 EIE crossing campuses across south India.

KTU D ZONE MEN'S FOOTBALL · 1ST PRIZE

Blessan Biju.

FISAT · December 2025

ANURANJ MEMORIAL · 1ST

Aaron Varghese *cricket champion.*

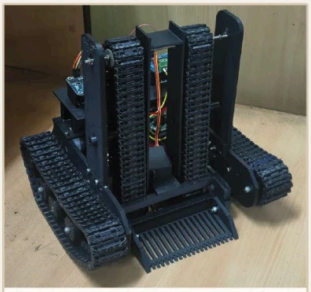
Took home the top spot at the **Anuranj Memorial Cricket Tournament** a hard-fought knockout campaign across multiple match days. A welcome trophy for Semester 6 sport.

FOUR CAMPUSES, FOUR EXPOS SEMESTER 6 ON TOUR

21 FEB 2026 · JCET

PALS innoWAH! Clusters

Anirudh M, **Govind Gopan** & **Vydhehy T M** at the JCET project expo.



ASME EFX · SJCET PALA

IAM3D Rover Competition

S5 & S6 fielded a rover entry under ASME EFx at SJCET Pala.

23-24 FEB 2026 · GITS

SHRISHTI-26

Dulfukkar Ali P B, **Fahad Yasin**, **Karthik M D**, **Anirudh M** 12th national project expo · Saint GITS.

20-21 FEB 2026 · MACE

Vajra Hackathon · SPECTRUM 5.6

Architha, **Blessan**, **Varun**, **Sushobh** at Vajra Hackathon.

The freshmen who refused to *wait.*

A balanced first-year achievement wall academic firsts, sport, culture, and enterprise kept on the same spread.

Mohammed Laqeen M V selected for the KTU University volleyball team.

VOLLEYBALL · KTU UNIVERSITY TEAM

Mohammed Laqeen M V.

Selected for the **KTU University volleyball team** after finishing **2nd in Interzone** and **1st in D Zone**. Also co-founded **@fee.sat_store** with **Muhammed Razal**.

Fida Fathima E N the first student in FISAT history to clear the Challenge Course in the very first semester.

ACADEMIC PATHBREAKER

First-semester challenge-course clearance.

Recognized here as the **first-ever challenge-course taker at FISAT** to pass in the very first semester a defining academic first for the batch.

Niranjan R · Group Dance wins with the FISAT Dance Team at MITS, Bharata Mata, Elims and Danza Season 5.

GROUP DANCE · CULTURAL ACHIEVER

Niranjan R.

With the **FISAT Dance Team**, Niranjan earned **first prize** in group dance at Muthoot Institute of Technology and Science and Bharata Mata College, **second prize** at Elims College of Arts and Science, and **fourth prize** at Danza Season 5.

MORE ON NEXT SPREAD

COMPLETE FIRST-YEAR ROSTER

Sports, culture & firsts.

Mohammed Laqeen M V KTU volleyball team; D-Zone 1st, Interzone 2nd.

Aryananda K S basketball: D-Zone runner-up; FISAT Trophy runner-up; Interzone.

Nivedya P S tug of war: KTU Interzone 1st.

Niranjan R FISAT Dance Team; group dance wins.

Muhammed Razal co-founded **@fee.sat_store** with Laqeen.

Ajith table tennis: D-Zone 2nd; Interzone at CET.

E S Sraddha table tennis: KTU D-Zone at FISAT.

Bharath R Battle of Bands 1st at YVIDH'26.

Fida Fathima E N first FISAT student to clear Challenge Course in S1.

Krishneandhu P K chess at MACE; maths quiz participant.

Niveditha K R kho-kho semifinalist; 4×100m relay 1st.

Nayan Kishore V K BHARATHAM'26 rap; *Reksha* on Spotify @nvkverse.

A first-year *photo wall.*

The full archive of freshman photographs that shaped this edition sport, stage, culture, and the candid moments of S1/S2 EIE.

Bharath R - Battle of Bands, 1st position.

Nayan Kishore V K - main-stage rap performance.

Krishneandhu P K - chess tournament and maths quiz.

Nayan Kishore V K - Reksha at BHARATHAM'26.

Muhammed Razal - co-founder, @fee.sat_store.

Aryananda K S - basketball, D-Zone runner-up.

Fida Fathima E N - challenge-course first.

Niranjan R - group dance team.

Ajith - table tennis recognition.

Aryananda K S - FISAT women's basketball squad.

Mohammed Laqeen M V - volleyball, KTU University team.

ARCHIVE · S1 / S2 EIE

Every uploaded frame, kept on the page.

A continuous record of how the batch showed up across the year on stage, in the gym, and out on field days.

Nivedya P S - tug of war, Interzone 1st prize.

Niranjan R - group dance performance.

E S Sraddha - table tennis, KTU D-Zone participation.

Freshman culture team - stage performance.

Niveditha K R - kho-kho semifinalist and 4x100m relay winner.

Where our students *go.*

02
DRDO · NPOL
Naval Physical & Oceanographic Lab

02
ISRO · VSSC
Aerospace electronics, Trivandrum

12+
CORE INDUSTRY
PLC · SCADA · IoT · Cryo controls

NPOL · DRDO · KOCHI

Akash Pradeep Kumar

S7 EIE · 6-MONTH PAID RESEARCH

Selected to the Naval Physical & Oceanographic Laboratory, DRDO Ministry of Defence, Govt. of India.

VSSC · ISRO · TRIVANDRUM

Naveen Jayaprakash & Vysakh C

AEROSPACE ELECTRONICS INTERNS

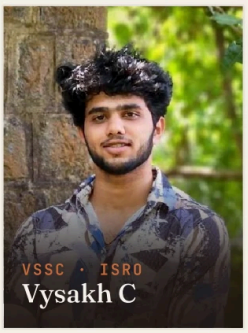
Placed at Vikram Sarabhai Space Centre, ISRO working at the heart of India's aerospace electronics programme.

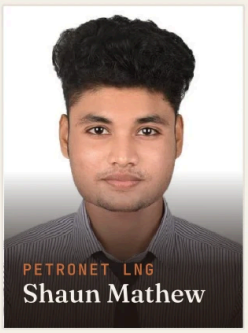
THE FULL 2025-26 ROSTER

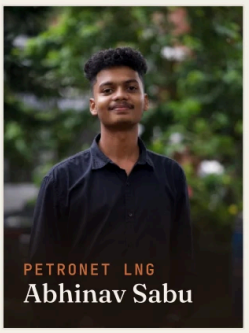
STUDENT	COMPANY & HUB	ROLE
Akash Pradeep Kumar · S7	NPOL · DRDO · Kochi	Research Intern · 6 mo · Paid
Naveen Jayaprakash · Vysakh C	VSSC · ISRO · Trivandrum	Aerospace electronics
Shaun Mathew · Abhinav Sabu · Annmary Benny	Petronet LNG · Kochi	Cryo-loop controls
Anna Baby · Parthiv A · Greeshma M	Izado Solutions · Kozhikode	Industrial IoT
Alen Manoj · Vishwam Sajeev	Regera Controls Pvt Ltd	PLC / SCADA
Loyd John J	He-Man Auto Robopark	Microcontrollers
Nikita Susan Binu	GOAT Robotics · Coimbatore	Robotics R&D
Geethika Sujith	Kalkitech · Bangalore	Grid telemetry
Gayathri Giridhar	Unique World Robotics India	Drone & automation
George Thomas · Nandakishor Kuruppath · Shahila T.I.	Afzar consultants LLC	UAE – Instrumentation

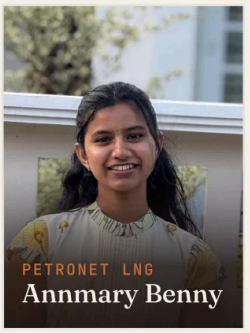
The faces *on assignment.*

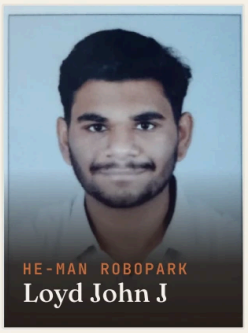
A portrait wall of EIE interns active across LNG, aerospace, robotics, automation and grid-tech teams.

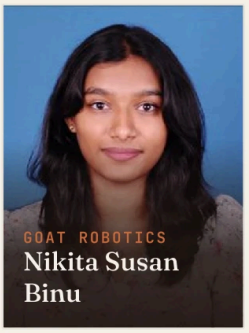


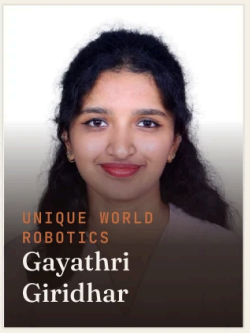






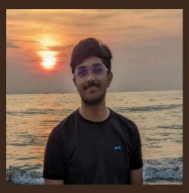






IZADO SOLUTIONS PVT LTD · KOZHIKODE

Anna Baby · Parthiv A · Greeshma M



Industrial IoT and embedded automation work at the Kozhikode Innovation Centre.

KALKITECH · BANGALORE

Geethika Sujith.

Working on grid telemetry & smart energy automation at Kalkitech a key engineering partner on Indias electrical-grid stack.

ADDITIONAL PLACEMENTS

George Thomas · Nandakishor Kuruppath · Shahila T.I.

Faculty *milestones.*

Dr. Sreevidya P · Assistant Professor, EIE · conferred her doctorate by Amrita Vishwa Vidyapeetham, Coimbatore, August 2025.

PHD CONFERRED · AUG 2025

Doctorate awarded to *Dr. Sreevidya P.*

Assistant Professor **Dr. Sreevidya P** was awarded her PhD by **Amrita Vishwa Vidyapeetham Deemed University, Coimbatore** in August 2025.

Enhancing personality & affect analysis by multimodal fusion with an application to recommender systems.

· Doctoral dissertation title

FACULTY ENGAGEMENT · NOV-DEC 2025

Ms. Raji P across entrepreneurship forums.

Ms. Raji P served as speaker for **Coffee with an Edupreneur** at Bharata Mata College of Commerce and Arts, Choondy, on 18 Dec 2025, and participated in the **3rd IIC Regional Meet** at Rajagiri Engineering College on 28 Nov 2025.

JOINT PATENTS · FISAT IPR CELL

01 **Dr. Abi P Mathew**
Patent published through the FISAT IPR Cell · 2025.

02 **Dr. S. Sundararajan**
Patent published through the FISAT IPR Cell · 2025.

03 **Dr. Sreevidya P**
Patent published through the FISAT IPR Cell · 2025.

A year of *milestones.*

Volume IX closes with the same spirit that shaped the year: faculty guidance, student initiative, and a department steadily building toward future-facing skills.

FACULTY MEMORIES FROM THE YEAR

IDEA association executive committee

EIE faculty and students at institutional forums and gatherings during the year.

AN ALIGNED INITIATIVE

Centre for Future Skills @ FISAT

A strategic initiative delivering internationally recognized, high-impact skill development for emerging technical domains.

The Centre for Future Skills @ FISAT an institutional skills initiative powered by NSDC and Ethnotech.

IDEA

INNOVATION • ELECTRONICS • INSTRUMENTATION

"This volume is not merely a record; it is a portrait of who we are becoming."

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DEPARTMENT OF ELECTRONICS & INSTRUMENTATION ENGINEERING

METRON